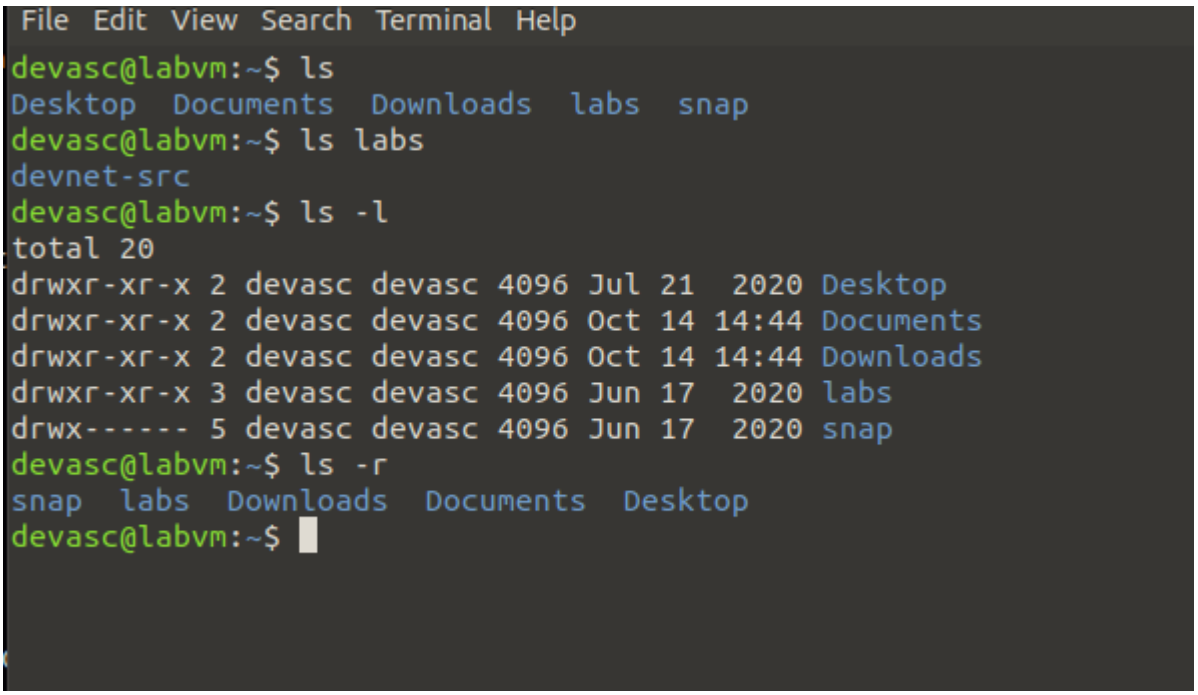
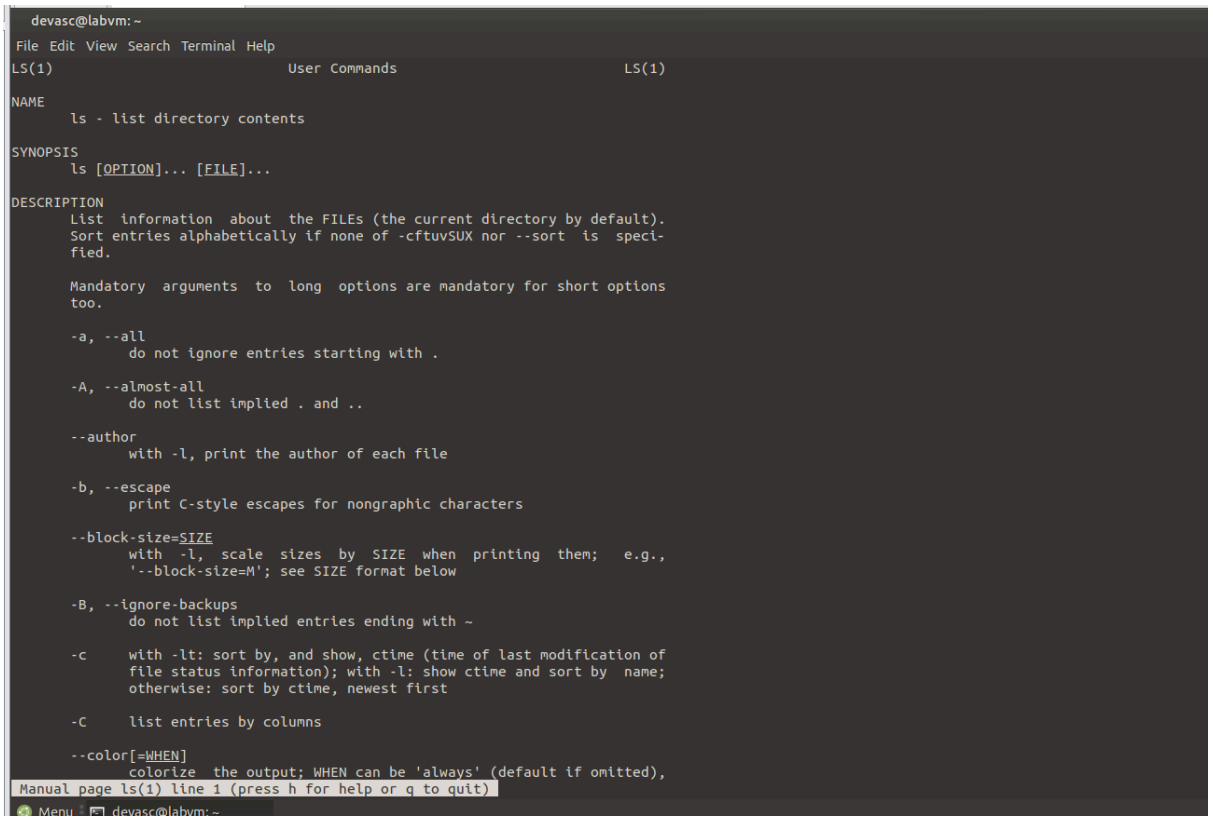

File Edit View Search Terminal Help
devasc@labvm:~$ ls
Desktop Documents Downloads labs snap
devasc@labvm:~$ ls labs
devnet-src
devasc@labvm:~$ ls -l
total 20
drwxr-xr-x 2 devasc devasc 4096 Jul 21 2020 Desktop
drwxr-xr-x 2 devasc devasc 4096 Oct 14 14:44 Documents
drwxr-xr-x 2 devasc devasc 4096 Oct 14 14:44 Downloads
drwxr-xr-x 3 devasc devasc 4096 Jun 17 2020 labs
drwx------ 5 devasc devasc 4096 Jun 17 2020 snap
devasc@labvm:~$ ls -r
snap labs Downloads Documents Desktop
devasc@labvm:~$


devasc@labvm: ~
File Edit View Search Terminal Help
LS(1) User Commands LS(1)
NAME
ls - list directory contents
SYNOPSIS
ls [OPTION]... [FILE]...
DESCRIPTION
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is speci-
fied.
Mandatory arguments to long options are mandatory for short options
too.
-a, --all
do not ignore entries starting with .
-A, --almost-all
do not list implied . and ..
--author
with -l, print the author of each file
-b, --escape
print C-style escapes for nongraphic characters
--block-size=SIZE
with -l, scale sizes by SIZE when printing them; e.g.,
'--block-size=M'; see SIZE format below
-B, --ignore-backups
do not list implied entries ending with ~
-c with -lt: sort by, and show, ctime (time of last modification of
file status information); with -l: show ctime and sort by name;
otherwise: sort by ctime, newest first
-C list entries by columns
--color[=WHEN]
colorize the output; WHEN can be 'always' (default if omitted),
Manual page ls(1) line 1 (press h for help or q to quit)
Menu devasc@labvm: ~

```
drwxr-xr-x 2 devasc devasc 4096 Jul 21  2020 Desktop
devasc@labvm:~$ man ls
devasc@labvm:~$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                  do not ignore entries starting with .
  -A, --almost-all           do not list implied . and ..
      --author               with -l, print the author of each file
  -b, --escape               print C-style escapes for nongraphic characters
      --block-size=SIZE      with -l, scale sizes by SIZE when printing them;
                               e.g., '--block-size=M'; see SIZE format below
  -B, --ignore-backups       do not list implied entries ending with ~
  -c                         with -lt: sort by, and show, ctime (time of last
                               modification of file status information);
                               with -l: show ctime and sort by name;
                               otherwise: sort by ctime, newest first
  -C                         list entries by columns
      --color[=WHEN]         colorize the output; WHEN can be 'always' (default
                               if omitted), 'auto', or 'never'; more info below
  -d, --directory            list directories themselves, not their contents
  -D, --dired                generate output designed for Emacs' dired mode
  -f                         do not sort, enable -aU, disable -ls --color
```

```
devasc@labvm:~$ pwd
/home/devasc
devasc@labvm:~$ cd Documents
devasc@labvm:~/Documents$ cd ..
devasc@labvm:~$ pwd
/home/devasc
devasc@labvm:~$
```

```
devasc@labvm:~$ pwd
/home/devasc
devasc@labvm:~$ cd Documents
devasc@labvm:~/Documents$ cd ..
devasc@labvm:~$ pwd
/home/devasc
devasc@labvm:~$ sudo apt-get update
Hit:1 http://archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:3 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:4 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3,611 kB]
Get:6 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [4,004 kB]
Get:7 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [891 kB]
Get:8 http://archive.ubuntu.com/ubuntu focal-updates/main i386 Packages [1,127 kB]
Get:9 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [518 kB]
Get:10 http://security.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [14.4 kB]
Get:11 http://archive.ubuntu.com/ubuntu focal-updates/main Translation-en [601 kB]
Get:12 http://security.ubuntu.com/ubuntu focal-security/restricted i386 Packages [42.8 kB]
Get:13 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [3,828 kB]
Get:14 http://archive.ubuntu.com/ubuntu focal-updates/main amd64 c-n-f Metadata [18.0 kB]
Get:15 http://archive.ubuntu.com/ubuntu focal-updates/restricted i386 Packages [44.3 kB]
Get:16 http://archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [3,984 kB]
Get:17 http://security.ubuntu.com/ubuntu focal-security/restricted Translation-en [527 kB]
Get:18 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 c-n-f Metadata [584 B]
Get:19 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [1,052 kB]
Get:20 http://archive.ubuntu.com/ubuntu focal-updates/restricted Translation-en [548 kB]
Get:21 http://archive.ubuntu.com/ubuntu focal-updates/restricted amd64 c-n-f Metadata [604 B]
Get:22 http://archive.ubuntu.com/ubuntu focal-updates/universe i386 Packages [832 kB]
Get:23 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1,275 kB]
Get:24 http://security.ubuntu.com/ubuntu focal-security/universe i386 Packages [705 kB]
Get:25 http://archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [302 kB]
Get:26 http://security.ubuntu.com/ubuntu focal-security/universe Translation-en [220 kB]
Get:27 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [29.3 kB]
Get:28 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [29.8 kB]
Get:29 http://archive.ubuntu.com/ubuntu focal-updates/multiverse i386 Packages [8,416 B]
Get:30 http://archive.ubuntu.com/ubuntu focal-updates/multiverse Translation-en [8,284 B]
Get:31 http://archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [688 B]
Get:32 http://archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [46.1 kB]
```

Part 3: Review File Management
Step 1: Review file permissions.
Step 2: Change file permissions and ownership.

```
devasc@labvm:~$ ls Desktop -l
total 36
-rwxr-xr-x 1 devasc devasc 1095 Jun 17  2020  chromium_chromium.desktop
-rwxr-xr-x 1 devasc devasc  228 Jun 17  2020  cisco-pt7.desktop
-rwxr-xr-x 1 devasc devasc  795 Jun 17  2020  code.desktop
-rwxrwxr-x 1 devasc devasc  319 Jul 21  2020 'DPI Scanling.desktop'
-rwxr-xr-x 1 devasc devasc  392 Jun 17  2020  drawio_drawio.desktop
-rwxr-xr-x 1 devasc devasc   99 Jun 17  2020  labs.desktop
-rwxr-xr-x 1 devasc devasc  284 Jun 17  2020  mate-keyboard.desktop
-rwxr-xr-x 1 devasc devasc  353 Jun 17  2020  postman_postman.desktop
-rwxr-xr-x 1 devasc devasc  254 Jun 17  2020  Terminal.desktop
devasc@labvm:~$ cd ~/Documents
devasc@labvm:~/Documents$ echo "ls ../Desktop" > myfile.sh
devasc@labvm:~/Documents$ cat myfile.sh
ls ../Desktop
devasc@labvm:~/Documents$ ./myfile.sh
bash: ./myfile.sh: Permission denied
devasc@labvm:~/Documents$
```

```
bash: ./myfile.sh: Permission denied
devasc@labvm:~/Documents$ chmod +x myfile.sh
devasc@labvm:~/Documents$ ./myfile.sh
 chromium_chromium.desktop   code.desktop           drawio_drawio.desktop   mate-keyboard.desktop     Terminal.desktop
 cisco-pt7.desktop          'DPI Scanling.desktop'   labs.desktop            postman_postman.desktop
devasc@labvm:~/Documents$
```

```
devasc@labvm:~/Documents$ sudo chown root myfile.sh
devasc@labvm:~/Documents$ ls -l myfile.sh
-rwxrwxr-x 1 root devasc 14 Oct 16 18:52 myfile.sh
devasc@labvm:~/Documents$
```

Step 3: Use the move files command.

```
devasc@labvm:~/Documents$ chmod +x myfile.sh
devasc@labvm:~/Documents$ ./myfile.sh
 chromium_chromium.desktop   code.desktop           drawio_drawio.desktop   mate-keyboard.desktop     Terminal.desktop
 cisco-pt7.desktop          'DPI Scanling.desktop'   labs.desktop            postman_postman.desktop
devasc@labvm:~/Documents$ sudo chown root myfile.sh
devasc@labvm:~/Documents$ ls -l myfile.sh
-rwxrwxr-x 1 root devasc 14 Oct 16 19:02 myfile.sh
devasc@labvm:~/Documents$ mv myfile.sh ~/Desktop/
devasc@labvm:~/Documents$ mv ~/Desktop/myfile.sh ~/Documents/
devasc@labvm:~/Documents$ mv myfile.sh myfile_renamed.sh
devasc@labvm:~/Documents$ cp myfile_renamed.sh myfile_renamed_and_copied.sh
devasc@labvm:~/Documents$ ls
myfile_renamed_and_copied.sh  myfile_renamed.sh
devasc@labvm:~/Documents$ rm myfile_renamed_and_copied.sh
devasc@labvm:~/Documents$ mv myfile.sh /home/devasc/Desktop/
mv: cannot stat 'myfile.sh': No such file or directory
devasc@labvm:~/Documents$ ls ../Desktop/
 chromium_chromium.desktop   code.desktop           drawio_drawio.desktop   mate-keyboard.desktop     Terminal.desktop
 cisco-pt7.desktop          'DPI Scanling.desktop'   labs.desktop            postman_postman.desktop
devasc@labvm:~/Documents$ ls -l
total 4
-rwxrwxr-x 1 root devasc 14 Oct 16 19:02 myfile_renamed.sh
devasc@labvm:~/Documents$ ls
myfile_renamed.sh
devasc@labvm:~/Documents$
Menu  devasc@labvm: ~/Doc...
```

Step 4: Use the copy files command.

```
devasc@labvm:~/Documents$ cp myfile_renamed.sh myfile_renamed_and_copied.sh
devasc@labvm:~/Documents$ ls
myfile_renamed_and_copied.sh  myfile_renamed.sh
```

Step 5: Use the remove files command.

```
devasc@labvm:~/Documents$ rm myfile_renamed_and_copied.sh
devasc@labvm:~/Documents$ ls
myfile_renamed.sh
```

Step 6: Use the standard output redirect.

```
devasc@labvm:~/Documents$ cd ~
devasc@labvm:~$ echo "Linux is AWESOME!" > linux.txt && \
> cat linux.txt > linux2.txt && \
> cat linux2.txt && \
> echo "I LOVE Linux!" >> linux2.txt && \
> cat linux2.txt && \
> echo "Linux is POWERFUL!" > linux.txt && \
> cat linux.txt
Linux is AWESOME!
Linux is AWESOME!
I LOVE Linux!
Linux is POWERFUL!
devasc@labvm:~$
```

Menu devasc@labvm: ~

Step 7: Use the vi text editor.

devasc@labvm: ~

File Edit View Search Terminal Help

```
Linux is Linux
I am AWESOME!
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
```

```
devasc@labvm:~$ cat linux3.txt
Linux is Linux
I am AWESOME!
devasc@labvm:~$
```

Part 4: Review Regular Expressions

```
devasc@labvm:~$ grep devasc /etc/passwd
devasc:x:900:900:DevNet Associate,,,:/home/devasc:/bin/bash
devasc@labvm:~$ grep root /etc/passwd
root:x:0:0:root:/root:/bin/bash
devasc@labvm:~$ grep '^root' /etc/passwd
root:x:0:0:root:/root:/bin/bash
devasc@labvm:~$ grep 'false$' /etc/passwd
tss:x:109:116:TPM software stack,,,:/var/lib/tpm:/bin/false
lightdm:x:110:120:Light Display Manager:/var/lib/lightdm:/bin/false
whoopsie:x:116:124::/nonexistent:/bin/false
speech-dispatcher:x:120:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
hplip:x:122:7:HPLIP system user,,,:/run/hplip:/bin/false
vboxadd:x:998:1::/var/run/vboxadd:/bin/false
devasc@labvm:~$ grep 'd.m' /etc/passwd
tcpdump:x:107:112::/nonexistent:/usr/sbin/nologin
systemd-coredump:x:999:999:systemd Core Dumper:/:/usr/sbin/nologin
devasc@labvm:~$ grep '[8-9]' /etc/passwd
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
sshd:x:108:65534::/run/sshd:/usr/sbin/nologin
devasc:x:900:900:DevNet Associate,,,:/home/devasc:/bin/bash
systemd-coredump:x:999:999:systemd Core Dumper:/:/usr/sbin/nologin
tss:x:109:116:TPM software stack,,,:/var/lib/tpm:/bin/false
cups-pk-helper:x:118:114:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
speech-dispatcher:x:120:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
saned:x:121:128::/var/lib/saned:/usr/sbin/nologin
colord:x:123:129:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
stunnel4:x:119:126::/var/run/stunnel4:/usr/sbin/nologin
vboxadd:x:998:1::/var/run/vboxadd:/bin/false
devasc@labvm:~$ grep '[,]' /etc/passwd
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
devasc:x:900:900:DevNet Associate,,,:/home/devasc:/bin/bash
tss:x:109:116:TPM software stack,,,:/var/lib/tpm:/bin/false
rtkit:x:111:122:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
avahi-autoipd:x:113:123:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:114:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
```

```
devasc@labvm:~$ grep '[,]' /etc/passwd
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
devasc:x:900:900:DevNet Associate,,,:/home/devasc:/bin/bash
tss:x:109:116:TPM software stack,,,:/var/lib/tpm:/bin/false
rtkit:x:111:122:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
avahi-autoipd:x:113:123:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:114:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
kernoops:x:115:65534:Kernel Oops Tracking Daemon,,,:/:/usr/sbin/nologin
avahi:x:117:125:Avahi mDNS daemon,,,:/var/run/avahi-daemon:/usr/sbin/nologin
cups-pk-helper:x:118:114:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
speech-dispatcher:x:120:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
hplip:x:122:7:HPLIP system user,,,:/run/hplip:/bin/false
colord:x:123:129:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
pulse:x:124:130:PulseAudio daemon,,,:/var/run/pulse:/usr/sbin/nologin
devasc@labvm:~$ grep 'new*' /etc/passwd
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:106::/nonexistent:/usr/sbin/nologin
_apt:x:105:65534::/nonexistent:/usr/sbin/nologin
tcpdump:x:107:112::/nonexistent:/usr/sbin/nologin
kernoops:x:115:65534:Kernel Oops Tracking Daemon,,,:/:/usr/sbin/nologin
whoopsie:x:116:124::/nonexistent:/bin/false
saned:x:121:128::/var/lib/saned:/usr/sbin/nologin
stunnel4:x:119:126::/var/run/stunnel4:/usr/sbin/nologin
devasc@labvm:~$
```

Part 5: Review System Administration

Step 1: Shut down the computer.

```
devasc@labvm:~$ date
Fri 16 Oct 2026 07:47:14 PM UTC
devasc@labvm:~$ sudo shutdown +1 "Come back soon!" && sudo shutdown -c
Shutdown scheduled for Fri 2026-10-16 19:48:20 UTC, use 'shutdown -c' to cancel.
```

Step 2: View and test the network configuration.

```
devasc@labvm:~$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:82:50:77 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.10/24 brd 192.168.1.255 scope global dynamic ens32
       valid_lft 81574sec preferred_lft 81574sec
    inet6 fe80::20c:29ff:fe82:5077/64 scope link
       valid_lft forever preferred_lft forever
3: dummy0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UNKNOWN group default qlen 1000
    link/ether 7e:e0:e1:1a:dc:4d brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.1/32 scope global dummy0
       valid_lft forever preferred_lft forever
    inet 192.0.2.2/32 scope global dummy0
       valid_lft forever preferred_lft forever
    inet 192.0.2.3/32 scope global dummy0
       valid_lft forever preferred_lft forever
    inet 192.0.2.4/32 scope global dummy0
       valid_lft forever preferred_lft forever
    inet 192.0.2.5/32 scope global dummy0
       valid_lft forever preferred_lft forever
    inet6 fe80::7ce0:e1ff:fe1a:dc4d/64 scope link
       valid_lft forever preferred_lft forever
devasc@labvm:~$
devasc@labvm:~$
```

```
devasc@labvm:~$  ip route
default via 192.168.1.1 dev ens32 proto dhcp src 192.168.1.10 metric 100
192.168.1.0/24 dev ens32 proto kernel scope link src 192.168.1.10
192.168.1.1 dev ens32 proto dhcp scope link src 192.168.1.10 metric 100
devasc@labvm:~$ ping -c 4 GATEWAY_IP
ping: GATEWAY_IP: Temporary failure in name resolution
devasc@labvm:~$ ping -c 4 www.cisco.com
PING www.cisco.com (23.197.160.120) 56(84) bytes of data.
64 bytes from www.cisco.com (23.197.160.120): icmp_seq=1 ttl=47 time=144 ms
64 bytes from www.cisco.com (23.197.160.120): icmp_seq=2 ttl=47 time=209 ms
64 bytes from www.cisco.com (23.197.160.120): icmp_seq=3 ttl=47 time=286 ms
64 bytes from www.cisco.com (23.197.160.120): icmp_seq=4 ttl=47 time=171 ms

--- www.cisco.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3001ms
rtt min/avg/max/mdev = 144.003/202.542/285.709/53.357 ms
```

Step 3: View Processes

```
devasc@labvm:~$ ps
    PID TTY          TIME CMD
   3401 pts/1    00:00:00 bash
   3588 pts/1    00:00:00 ps
devasc@labvm:~$ ps -e
    PID TTY          TIME CMD
      1 ?        00:00:06 systemd
      2 ?        00:00:00 kthreadd
      3 ?        00:00:00 rcu_gp
      4 ?        00:00:00 rcu_par_gp
      6 ?        00:00:00 kworker/0:0H-kblockd
      9 ?        00:00:00 mm_percpu_wq
     10 ?        00:00:00 ksoftirqd/0
     11 ?        00:00:04 rcu_sched
     12 ?        00:00:00 migration/0
     13 ?        00:00:00 idle_inject/0
     14 ?        00:00:00 cpuhp/0
     15 ?        00:00:00 cpuhp/1
     16 ?        00:00:00 idle_inject/1
     17 ?        00:00:00 migration/1
     18 ?        00:00:00 ksoftirqd/1
     20 ?        00:00:00 kworker/1:0H-kblockd
     21 ?        00:00:00 kdevtmpfs
     22 ?        00:00:00 netns
     23 ?        00:00:00 rcu_tasks_kthre
     24 ?        00:00:00 kauditd
     25 ?        00:00:00 khungtaskd
     26 ?        00:00:00 oom_reaper
     27 ?        00:00:00 writeback
     28 ?        00:00:00 kcompactd0
     29 ?        00:00:00 ksmd
     30 ?        00:00:00 khugepaged
    123 ?        00:00:00 kintegrityd
    124 ?        00:00:00 kblockd
    125 ?        00:00:00 blkcg_punt_bio
    126 ?        00:00:00 tpm_dev_wq
    127 ?        00:00:00 ata_sff
    128 ?        00:00:00 md
    129 ?        00:00:00 edac-poller
    130 ?        00:00:00 devfreq_wq
    131 ?        00:00:00 watchdogd
    134 ?        00:00:00 kswapd0
```

```
File  Edit  View  Search  Terminal  Help
   3523 ?        00:00:00 kworker/u4:0-events_power_efficient
   3535 ?        00:00:00 kworker/u4:1-events_unbound
   3562 ?        00:00:00 kworker/u4:2-events_unbound
   3565 ?        00:00:00 kworker/0:2-events
   3583 ?        00:00:00 kworker/0:3
   3584 ?        00:00:00 kworker/1:0-events
   3585 ?        00:00:00 kworker/1:2-events
   3587 ?        00:00:00 kworker/1:4
   3589 pts/1    00:00:00 ps
devasc@labvm:~$ ps -e | more
    PID TTY          TIME CMD
      1 ?        00:00:06 systemd
      2 ?        00:00:00 kthreadd
      3 ?        00:00:00 rcu_gp
      4 ?        00:00:00 rcu_par_gp
      6 ?        00:00:00 kworker/0:0H-kblockd
      9 ?        00:00:00 mm_percpu_wq
     10 ?        00:00:00 ksoftirqd/0
     11 ?        00:00:04 rcu_sched
     12 ?        00:00:00 migration/0
     13 ?        00:00:00 idle_inject/0
     14 ?        00:00:00 cpuhp/0
     15 ?        00:00:00 cpuhp/1
     16 ?        00:00:00 idle_inject/1
     17 ?        00:00:00 migration/1
     18 ?        00:00:00 ksoftirqd/1
     20 ?        00:00:00 kworker/1:0H-kblockd
     21 ?        00:00:00 kdevtmpfs
     22 ?        00:00:00 netns
     23 ?        00:00:00 rcu_tasks_kthre
     24 ?        00:00:00 kauditd
     25 ?        00:00:00 khungtaskd
     26 ?        00:00:00 oom_reaper
     27 ?        00:00:00 writeback
     28 ?        00:00:00 kcompactd0
     29 ?        00:00:00 ksmd
     30 ?        00:00:00 khugepaged
    123 ?        00:00:00 kintegrityd
    124 ?        00:00:00 kblockd
    125 ?        00:00:00 blkcg_punt_bio
    126 ?        00:00:00 tpm_dev_wq
    127 ?        00:00:00 ata_sff
    128 ?        00:00:00 md
    129 ?        00:00:00 edac-poller
```

```
devasc@labvm:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 18:26 ?        00:00:06 /sbin/init
root           2       0  0 18:26 ?        00:00:00 [kthreadd]
root           3       2  0 18:26 ?        00:00:00 [rcu_gp]
root           4       2  0 18:26 ?        00:00:00 [rcu_par_gp]
root           6       2  0 18:26 ?        00:00:00 [kworker/0:0H-kblockd]
root           9       2  0 18:26 ?        00:00:00 [mm_percpu_wq]
root          10       2  0 18:26 ?        00:00:00 [ksoftirqd/0]
root          11       2  0 18:26 ?        00:00:06 [rcu_sched]
root          12       2  0 18:26 ?        00:00:00 [migration/0]
root          13       2  0 18:26 ?        00:00:00 [idle_inject/0]
root          14       2  0 18:26 ?        00:00:00 [cpuhp/0]
root          15       2  0 18:26 ?        00:00:00 [cpuhp/1]
root          16       2  0 18:26 ?        00:00:00 [idle_inject/1]
root          17       2  0 18:26 ?        00:00:00 [migration/1]
root          18       2  0 18:26 ?        00:00:00 [ksoftirqd/1]
root          20       2  0 18:26 ?        00:00:00 [kworker/1:0H-kblockd]
root          21       2  0 18:26 ?        00:00:00 [kdevtmpfs]
root          22       2  0 18:26 ?        00:00:00 [netns]
root          23       2  0 18:26 ?        00:00:00 [rcu_tasks_kthre]
root          24       2  0 18:26 ?        00:00:00 [kauditd]
root          25       2  0 18:26 ?        00:00:00 [khungtaskd]
root          26       2  0 18:26 ?        00:00:00 [oom_reaper]
root          27       2  0 18:26 ?        00:00:00 [writeback]
root          28       2  0 18:26 ?        00:00:00 [kcompactd0]
root          29       2  0 18:26 ?        00:00:00 [ksmd]
root          30       2  0 18:26 ?        00:00:00 [khugepaged]
root         123       2  0 18:26 ?        00:00:00 [kintegrityd]
root         124       2  0 18:26 ?        00:00:00 [kblockd]
root         125       2  0 18:26 ?        00:00:00 [blkcg_punt_bio]
root         126       2  0 18:26 ?        00:00:00 [tpm_dev_wq]
root         127       2  0 18:26 ?        00:00:00 [ata_sff]
root         128       2  0 18:26 ?        00:00:00 [md]
root         129       2  0 18:26 ?        00:00:00 [edac-poller]
root         130       2  0 18:26 ?        00:00:00 [devfreq_wq]
root         131       2  0 18:26 ?        00:00:00 [watchdogd]
root         134       2  0 18:26 ?        00:00:00 [kswapd0]
root         135       2  0 18:26 ?        00:00:00 [ecryptfs-kthrea]
root         138       2  0 18:26 ?        00:00:00 [kthrotld]
root         139       2  0 18:26 ?        00:00:00 [irq/24-pciehp]
root         140       2  0 18:26 ?        00:00:00 [irq/25-pciehp]
root         141       2  0 18:26 ?        00:00:00 [irq/26-pciehp]
```

Step 4: Manage packages.

```
devasc@labvm:~$ sudo apt-get update
Hit:1 http://security.ubuntu.com/ubuntu focal-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu focal InRelease
Hit:3 http://archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu focal-backports InRelease
Reading package lists... Done
devasc@labvm:~$ apt-cache search speedtest
speedtest-cli - Command line interface for testing internet bandwidth using speedtest.net
devasc@labvm:~$ sudo apt-get install speedtest-cli
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  speedtest-cli
0 upgraded, 1 newly installed, 0 to remove and 675 not upgraded.
Need to get 24.0 kB of archives.
After this operation, 106 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 speedtest-cli all 2.1.2-2ubuntu0.20.04.1 [24.0 kB]
Fetched 24.0 kB in 3s (9,423 B/s)
Selecting previously unselected package speedtest-cli.
(Reading database ... 211981 files and directories currently installed.)
Preparing to unpack .../speedtest-cli_2.1.2-2ubuntu0.20.04.1_all.deb ...
Unpacking speedtest-cli (2.1.2-2ubuntu0.20.04.1) ...
Setting up speedtest-cli (2.1.2-2ubuntu0.20.04.1) ...
Processing triggers for man-db (2.9.1-1) ...
devasc@labvm:~$ speedtest-cli
Retrieving speedtest.net configuration...
Testing from OJSC Kyrgyztelecom (212.241.4.93)...
Retrieving speedtest.net server list...
Selecting best server based on ping...
Hosted by MegaCom (Bishkek) [156.16 km]: 35.397 ms
Testing download speed................................................................................
Download: 5.63 Mbit/s
Testing upload speed................................................................................................
Upload: 1.16 Mbit/s
```




```
devasc@labvm:~$ sudo apt-get purge speedtest-cli
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages will be REMOVED:
  speedtest-cli*
0 upgraded, 0 newly installed, 1 to remove and 27 not upgraded.
After this operation, 106 kB disk space will be freed.
Do you want to continue? [Y/n] y
(Reading database ... 213214 files and directories currently installed.)
Removing speedtest-cli (2.1.2-2ubuntu0.20.04.1) ...
Processing triggers for man-db (2.9.1-1) ...
devasc@labvm:~$
```

Step 5: Update Passwords

```
devasc@labvm:~$ passwd
Changing password for devasc.
Current password:
passwd: Authentication token manipulation error
passwd: password unchanged
devasc@labvm:~$ sudo touch /tmp/testfile && echo OK
OK
devasc@labvm:~$ passwd -S
devasc P 06/17/2020 0 99999 7 -1
devasc@labvm:~$ man passwd
devasc@labvm:~$ sudo passwd devasc
New password:
Retype new password:
passwd: password updated successfully
devasc@labvm:~$
```

